Seminars 1 and 2: Sinusoids and Complex Exponentials Questions and Answers

1. Review of sine and cosine

Make carefully labelled sketches, marking amplitude, period, maxima, minima, and zero crossings.

(a) Question. Sketch $x_1(\theta) = \cos\theta$ for $0 \leq \theta \leq 6\pi$.

(a) Answer. $A = 1, T = 2\pi$; maxima at $0, 2\pi, 4\pi, 6\pi$, minima at $\pi, 3\pi, 5\pi$, zeros at $\pi/2 + k\pi$.

(b) Question. Sketch $x_2(t) = 2\cos(0.4\pi t)$ over three periods.

(b) Answer. $A = 2, T_0 = 5$ s; plot over $0 \leq t \leq 15$ s.

(c) Question. Sketch $x_3(t) = 2\cos(0.4\pi t + \pi/2)$ over three periods.

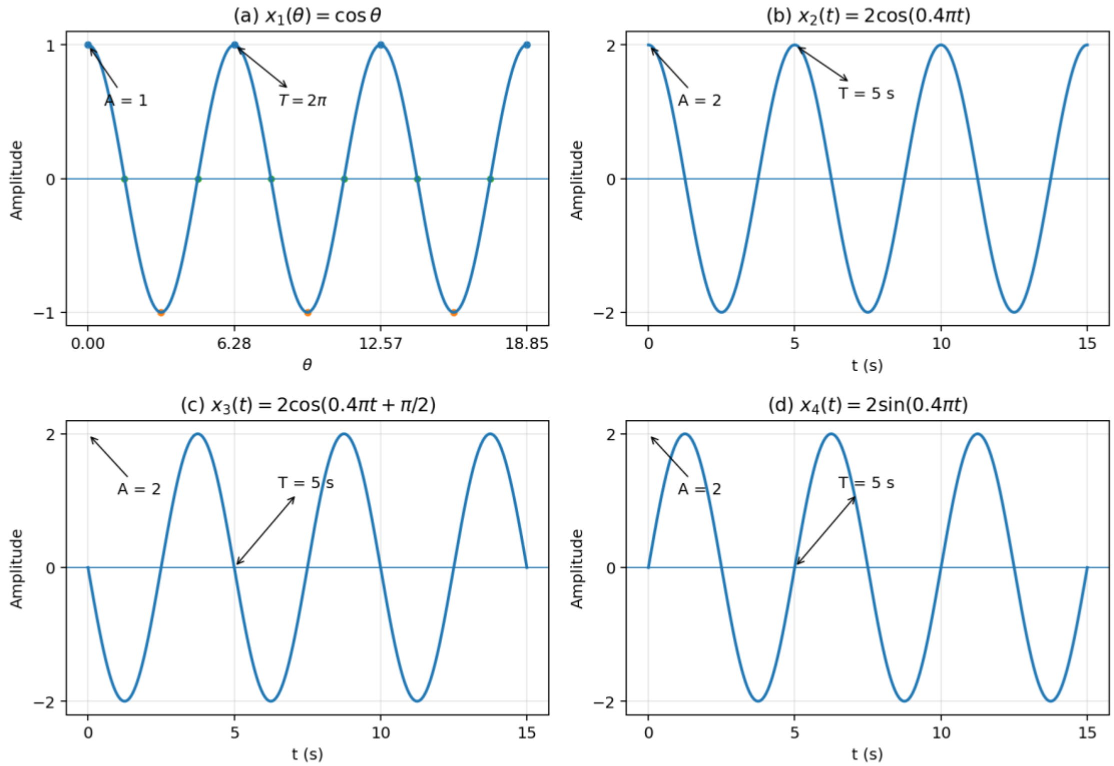
(c) Answer. $A = 2, T_0 = 5$ s, and $x_3(t) = -2\sin(0.4\pi t)$.

(d) Question. Sketch $x_4(t) = 2\sin(0.4\pi t)$ over three periods.

(d) Answer. $A = 2, T_0 = 5$ s.

(e) Question. Explain the relation between the graphs in parts (c) and (d).

(e) Answer. $x_3(t) = -x_4(t)$.



Labelled sine and cosine plots for Exercise 1

2. Reading sinusoidal parameters from a graph

The graph represents $x(t) = A\cos(\omega_0 t + \phi)$, with maximum 3, minimum -3, successive maxima separated by 2 s, and $x(0) = 1.5$ with positive slope.

(a) Question. Determine the amplitude A.

(a) Answer. $A = 3$.

(b) Question. Determine the period T_0.

(b) Answer. $T_0 = 2$ s.

(c) Question. Determine the cyclic frequency f_0.

(c) Answer. $f_0 = 0.5$ Hz.

(d) Question. Determine the radian frequency ω_0.

(d) Answer. $\omega_0 = \pi$ rad/s.

(e) Question. Determine one valid phase ϕ in $-\pi < \phi \leq \pi$.

(e) Answer. $\phi = -\pi/3$, so $x(t) = 3\cos(\pi t - \pi/3)$.

3. Frequency and period

For each signal determine A, ω_0, f_0, T_0, and ϕ, with $-\pi < \phi \leq \pi$. **(a) Question.** $x_1(t) = 5\cos(6\pi t - \pi/4)$.

(a) Answer. $A = 5$, $\omega_0 = 6\pi$, $f_0 = 3$ Hz, $T_0 = 1/3$ s, $\phi = -\pi/4$.

(b) Question. $x_2(t) = 3\sin(20\pi t + \pi/6)$; first rewrite it as a cosine.

(b) Answer. $x_2(t) = 3\cos(20\pi t - \pi/3)$; $A = 3$, $\omega_0 = 20\pi$, $f_0 = 10$ Hz, $T_0 = 0.1$ s, $\phi = -\pi/3$.

(c) Question. $x_3(t) = -4\cos(8t + 5\pi/3)$, using a positive amplitude.

(c) Answer. $x_3(t) = 4\cos(8t + 2\pi/3)$; $A = 4$, $\omega_0 = 8$, $f_0 = 4/\pi$ Hz, $T_0 = \pi/4$ s, $\phi = 2\pi/3$.

(d) Question. $x_4(t) = 2\cos(120\pi t + 7\pi/2)$.

(d) Answer. $A = 2$, $\omega_0 = 120\pi$, $f_0 = 60$ Hz, $T_0 = 1/60$ s, $\phi = -\pi/2$.

4. Phase and time shift

(a) Question. Derive the relation between ϕ and t_1 in $A\cos(\omega_0 t + \phi) = A\cos[\omega_0(t - t_1)]$.

(a) Answer. $\phi = -\omega_0 t_1$, $t_1 = -\phi/\omega_0$.

(b) Question. For $x(t) = 4\cos(10\pi t - 3\pi/4)$, determine the smallest positive delay.

(b) Answer. $t_1 = 3/40 \text{ s} = 0.075 \text{ s}$.

(c) Question. Rewrite $x(t) = 3\sin(12\pi t)$ in delayed-cosine form.

(c) Answer. $x(t) = 3\cos[12\pi(t - 1/24)]$.

(d) Question. Explain why the same sinusoid has infinitely many equivalent phases and time shifts.

(d) Answer. $\phi \equiv \phi + 2\pi k$ and $t_1 \equiv t_1 + kT_0$.

5. Maxima, minima, and zero crossings

Consider $x(t) = 6\cos(8\pi t + \pi/3)$. **(a) Question.** Find the first maximum at or after $t = 0$.

(a) Answer. $t_{\text{max}} = 5/24$ s.

(b) Question. Find the first minimum at or after $t = 0$.

(b) Answer. $t_{\text{min}} = 1/12$ s.

(c) Question. Find the first two positive zero crossings.

(c) Answer. $t = 1/48$ s and $t = 7/48$ s.

(d) Question. Verify by sketching two periods.

(d) Answer. $T_0 = 1/4$ s; sketch over $0 \le t \le 0.5$ s.

6. Sampling a continuous-time sinusoid

Let $x(t) = 20\cos(80\pi t - 0.4\pi)$. **(a) Question.** Write the sampled sequence for $T_s = 0.005$ s.

(a) Answer. $x[n] = 20\cos(0.4\pi n - 0.4\pi)$.

(b) Question. How many samples are taken per period?

(b) Answer. 5 samples per period.

(c) Question. Calculate $x[n]$ for $n = 0,1,2,3,4$.

(c) Answer. $\{6.1803, 20, 6.1803, -16.1803, -16.1803\}$.

(d) Question. Repeat parts (a)-(c) for $T_s = 0.0025$ s and state which plot is smoother.

(d) Answer. $x[n] = 20\cos(0.2\pi n - 0.4\pi)$; 10 samples/period; first five values $\{6.1803, 16.1803, 20, 16.1803, 6.1803\}$; this plot is smoother.

(e) Question. Plot the continuous signal and both sampled sequences over two periods.

(e) Answer. Plot over $0 \leq t \leq 0.05$ s; the $T_s = 0.0025$ s samples are denser.

7. Complex-number review for sinusoidal analysis

Convert between Cartesian and polar form and plot each number in the complex plane.

(a) Question. $z_1 = 3 + j4$.

(a) Answer. $z_1 = 5e^{j0.9273}$.

(b) Question. $z_2 = -2 + j2\sqrt{3}$.

(b) Answer. $z_2 = 4e^{j2\pi/3}$.

(c) Question. $z_3 = 5e^{-j\pi/6}$.

(c) Answer. $z_3 = 5\sqrt{3}/2 - j5/2$.

(d) Question. $z_4 = 2e^{j5\pi/4}$.

(d) Answer. $z_4 = -\sqrt{2} - j\sqrt{2}$.

(e) Question. Compute $z_1 z_3$.

(e) Answer. $z_1 z_3 \approx 22.990 + j9.820$.

(f) Question. Compute z_1/z_3.

(f) Answer. $z_1/z_3 \approx 0.1196 + j0.9928$.

8. Euler's formula and inverse Euler formulas

Start from $e^{j\theta} = \cos\theta + j\sin\theta$. **(a) Question.** Derive $e^{-j\theta}$.

(a) Answer. $e^{-j\theta} = \cos\theta - j\sin\theta$.

(b) Question. Derive the inverse Euler formula for cosine.

(b) Answer. $\cos\theta = \left(e^{j\theta} + e^{-j\theta}\right)/2$.

(c) Question. Derive the inverse Euler formula for sine.

(c) Answer. $\sin\theta = \left(e^{j\theta} - e^{-j\theta}\right)/(2j)$.

(d) Question. Express $3\cos(5t - \pi/4)$ as two complex exponentials.

(d) Answer. $3\cos(5t - \pi/4) = 3/2\, e^{j(5t-\pi/4)} + 3/2\, e^{-j(5t-\pi/4)}$.

9. Rotating complex exponentials

Consider $z(t) = 2e^{j(\pi t+\pi/4)}$. **(a) Question.** Determine magnitude and phase at $t = 0$.

(a) Answer. Magnitude 2; phase $\pi/4$.

(b) Question. Determine the direction of rotation.

(b) Answer. Counter-clockwise.

(c) Question. Determine the period.

(c) Answer. $T = 2$ s.

(d) Question. Calculate $z(t)$ at $t = 0{,}0.25{,}0.5{,}0.75{,}1$ s.

(d) Answer. $\{\sqrt{2} + j\sqrt{2},\ 2j,\ -\sqrt{2} + j\sqrt{2},\ -2,\ -\sqrt{2} - j\sqrt{2}\}$.

(e) Question. Sketch the real and imaginary parts over one period.

(e) Answer. $\Re\{z(t)\} = 2\cos(\pi t + \pi/4)$, $\Im\{z(t)\} = 2\sin(\pi t + \pi/4)$.

10. Representing a real sinusoid with a phasor

For each signal identify the phasor X and a complex signal $z(t)$ such that $x(t) = \Re\{z(t)\}$. **(a) Question.** $x_1(t) = 4\cos(100\pi t - \pi/3)$.

(a) Answer. $X_1 = 4e^{-j\pi/3}$, $z_1(t) = 4e^{-j\pi/3}e^{j100\pi t}$.

(b) Question. $x_2(t) = 3\sin(40\pi t + \pi/4)$.

(b) Answer. $X_2 = 3e^{-j\pi/4}$, $z_2(t) = 3e^{-j\pi/4}e^{j40\pi t}$.

(c) Question. $x_3(t) = -2\cos(60\pi t - \pi/6)$, using a positive magnitude.

(c) Answer. $X_3 = 2e^{j5\pi/6}$, $z_3(t) = 2e^{j5\pi/6}e^{j60\pi t}$.

(d) Question. Plot the three phasors.

(d) Answer. Angles $-\pi/3, -\pi/4, 5\pi/6$ with magnitudes 4,3,2.

11. Addition of two same-frequency sinusoids

Combine $x(t) = 2\cos(200\pi t + \pi/3) + 2\cos(200\pi t - 3\pi/4)$ into one cosine. **(a) Question.** Draw the two original phasors.

(a) Answer. $X_1 = 2e^{j\pi/3}$, $X_2 = 2e^{-j3\pi/4}$.

(b) Question. Add them in Cartesian form.

(b) Answer. $X = (1 - \sqrt{2}) + j(\sqrt{3} - \sqrt{2})$.

(c) Question. Convert the sum to polar form.

(c) Answer. $X \approx 0.5221e^{j19\pi/24}$.

(d) Question. State the resulting sinusoid.

(d) Answer. $x(t) \approx 0.5221\cos(200\pi t + 19\pi/24)$.

(e) Question. Verify the result at $t = 0$.

(e) Answer. Both forms give $1 - \sqrt{2} \approx -0.4142$.

12. Addition of three same-frequency sinusoids

Combine $x(t) = 2\cos(300\pi t) + 3\cos(300\pi t + \pi/2) + 4\cos(300\pi t - \pi)$ into one cosine. **(a) Question.** Draw the three phasors.

(a) Answer. $X_1 = 2, X_2 = 3j, X_3 = -4$.

(b) Question. Add them graphically and algebraically.

(b) Answer. $X = -2 + 3j$.

(c) Question. Express the result as $A\cos(300\pi t + \phi)$.

(c) Answer. $x(t) = \sqrt{13}\cos(300\pi t + 2.1588)$.

(d) Question. Explain what changes if one term has a different frequency.

(d) Answer. Different-frequency terms cannot generally be reduced to one sinusoid.

13. General two-phasor addition identity

Let $X_3 = e^{j\alpha} + e^{j\beta}$. **(a) Question.** Show that $X_3 = 2\cos[(\alpha - \beta)/2]e^{j(\alpha+\beta)/2}$.

(a) Answer. $X_3 = 2\cos[(\alpha - \beta)/2]e^{j(\alpha+\beta)/2}$.

(b) Question. Interpret magnitude and angle geometrically.

(b) Answer. Magnitude $2|\cos[(\alpha - \beta)/2]|$; angle along the bisector $(\alpha + \beta)/2$.

(c) Question. What happens when $\alpha - \beta = \pi$?

(c) Answer. $X_3 = 0$.

(d) Question. What happens when $\alpha = \beta$?

(d) Answer. $X_3 = 2e^{j\alpha}$.

14. Discrete-time complex exponential

Consider $x[n] = 2e^{j(\pi n/4+\pi/6)}$. **(a) Question.** State magnitude, normalized radian frequency, and initial phase.

(a) Answer. 2, $\pi/4$ rad/sample, $\pi/6$.

(b) Question. Determine the fundamental period in samples.

(b) Answer. $N_0 = 8$.

(c) Question. Plot $\Re\{x[n]\}$ and $\Im\{x[n]\}$ versus n.

(c) Answer. $\Re\{x[n]\} = 2\cos(\pi n/4 + \pi/6)$; $\Im\{x[n]\} = 2\sin(\pi n/4 + \pi/6)$.

(d) Question. Plot the complex samples and explain why they repeat.

(d) Answer. Circle of radius 2; repeats every 8 samples.

15. Integrated computational exercise

Using Python, analyse $x(t) = 5\cos(2\pi\, 110t + \pi/3) + 4\cos(2\pi\, 110t - \pi/4)$. **(a) Question.** Plot both components and their sum over three periods.

(a) Answer. Use $0 \le t \le 3/110$ s.

(b) Question. Draw both phasors and their vector sum.

(b) Answer. $X = 5.3284 + j1.5017$.

(c) Question. Compute the equivalent amplitude and phase.

(c) Answer. $A \approx 5.5360$, $\phi \approx 0.2747$ rad.

(d) Question. Write and plot the equivalent single sinusoid.

(d) Answer. $x(t) = 5.5360\cos(2\pi\, 110t + 0.2747)$.

(e) Question. Choose a sampling rate giving at least 20 samples per period.

(e) Answer. $f_s \ge 2200$ samples/s; choose $f_s = 2200$ samples/s.